SLIDE 1

YOUR DATABASE PASSWORD IS IN YOUR CODE!

DEVELOPER → GITHUB REPOSITORY →

```
application.yml
spring:
  datasource:
    url: jdbc:mysql://mydb
    username: admin
    password: Admin@123
```

EXPOSED SECRET!

WHAT IF THIS REPOSITORY LEAKS?

↓

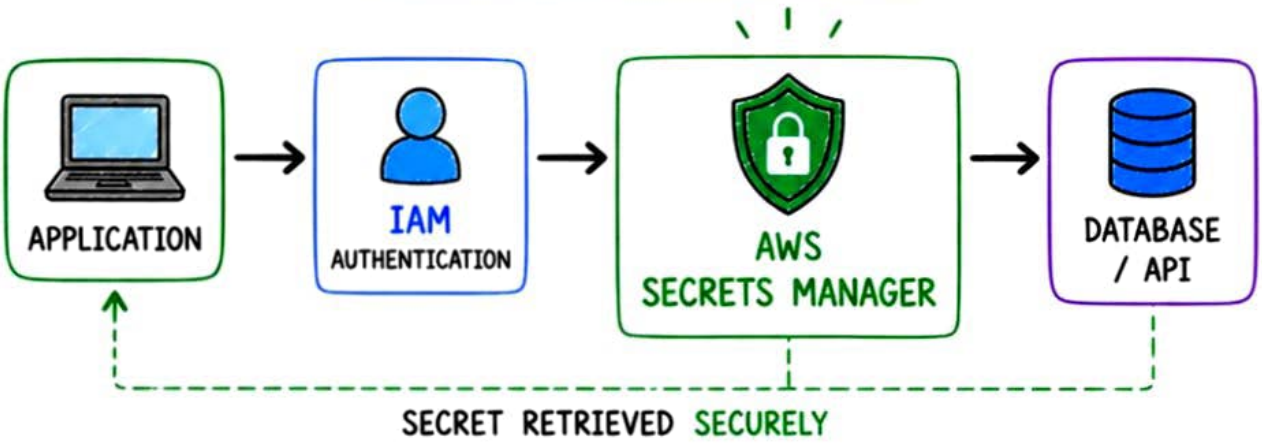
AWS SECRETS MANAGER

Secrets belong in a secret store – NOT in source code.

SLIDE 2

WHAT IS AWS SECRETS MANAGER?

A managed AWS service for securely storing, retrieving and rotating secrets.



EXAMPLES OF SECRETS

- Database credentials (user & password)
- API keys
- Application credentials
- OAuth tokens
- Other sensitive configuration

KEY BENEFITS

- [x] Centralized & secure storage
- [x] Fine-grained access with IAM
- [x] Automatic secret rotation
- [x] Encryption with AWS KMS
- [x] Audit & monitoring with CloudTrail

APPLICATION RETRIEVES THE SECRET AT RUNTIME – NOT AT BUILD TIME.

SLIDE 3

HOW DOES IT WORK?

YOUR APPLICATION GETS A SECRET

1. 

APPLICATION
Needs the secret at runtime

2. 

IAM ROLE
Application authenticates using IAM role

3. 

AWS SECRETS MANAGER
Secret is fetched securely

4. 

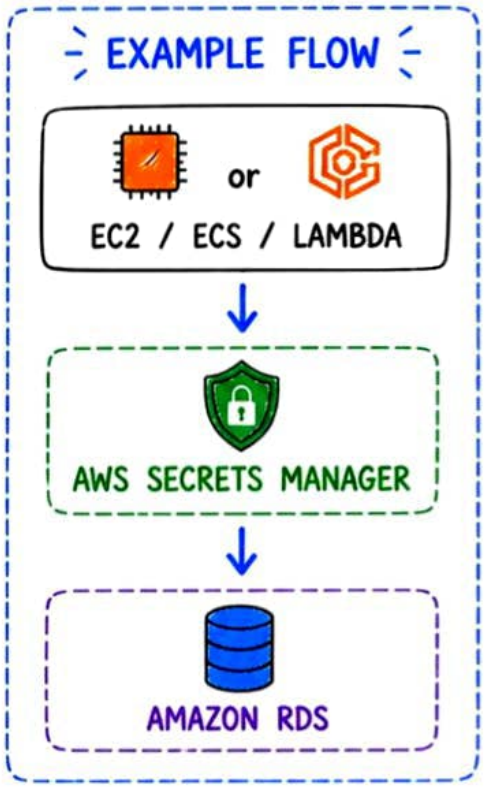
AWS KMS
Secret is encrypted using AWS KMS

5. 

SECRET RETURNED
Decrypted & returned to the application

6. 

CONNECT TO BACKEND
Application uses the secret to connect to RDS / API



KEY IDEA

- [x] No hardcoded credentials
- [x] No secrets in code
- [x] Retrieve securely at runtime

SECURE, SIMPLE & AUTOMATED WAY TO MANAGE YOUR SECRETS.

SLIDE 4

AUTOMATIC SECRET ROTATION

Rotate secrets automatically & keep your application running without interruption!

ROTATION HAPPENS AUTOMATICALLY

1. Generate / Update Credential
2. Update the database — RDS
3. Store the new secret — SECRETS MANAGER
4. Applications retrieve the current value

APPLICATION — Always uses the latest secret

Secrets Manager rotation Lambda function updates the secret in the target service (like RDS) and keeps your application running with zero downtime.

BENEFITS

- Automatically rotate secrets
- Improve security posture
- No manual intervention
- Reduce risk of credential leaks

ROTATE OFTEN. STAY SECURE. STAY AHEAD.

SLIDE 5

SECRETS MANAGER VS PARAMETER STORE

WHICH ONE SHOULD YOU USE?

FEATURE / CAPABILITY	AWS SECRETS MANAGER	AWS PARAMETER STORE
Primary purpose	Store, manage & rotate secrets	Store configuration & parameters
Store secrets	Yes	Yes (SecureString)
Automatic rotation	Built-in rotation	Manual / DIY
Database credential rotation	Native support (RDS, etc.)	DIY with Lambda
Cost	Paid service	Standard tier is low-cost
Best for	Secrets lifecycle, rotation & auditing	App configs, feature flags, non-rotating parameters
Scaling & operations	Managed, highly scalable, minimal ops	Highly scalable, serverless
Integration with AWS services	Deep integration with RDS, Redshift, etc.	Works across many AWS services

USE CASE TIP

Use SECRETS MANAGER when you need secret management + rotation. → Example: RDS password, API keys, DB credentials

Use PARAMETER STORE when you need config management or simple parameters. → Example: App settings, feature flags, endpoints

Right tool. Right use case. Better Security. Lower Cost.

THAT'S CLOUD SMARTNESS!

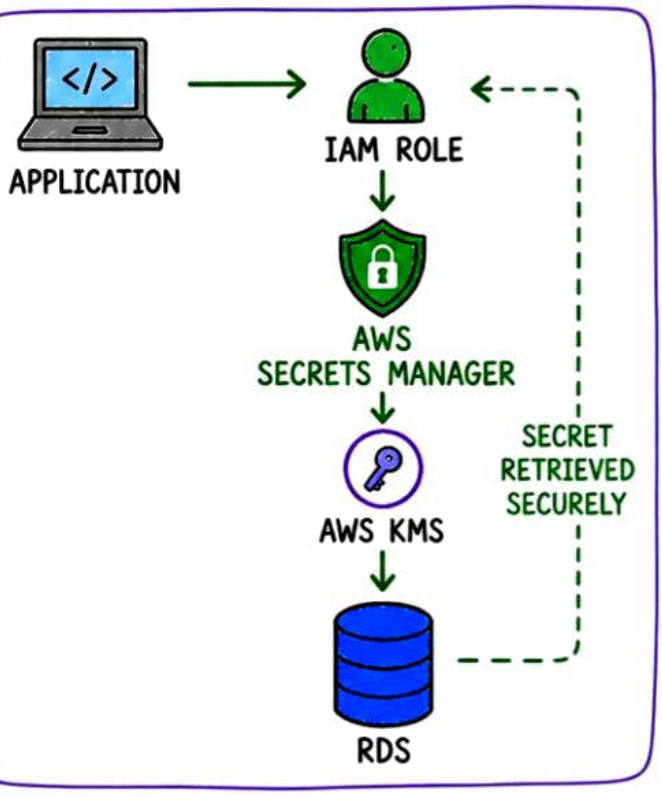
SLIDE 6

INTERVIEW CHEAT SHEET

HOW WOULD YOU STORE AN RDS PASSWORD IN PRODUCTION?

STRONG ANSWER FRAMEWORK

1. Store the password in AWS SECRETS MANAGER.
2. Encrypt the secret using AWS KMS.
3. Grant access using IAM role (least privilege).
4. Application retrieves secret at runtime.
5. Enable automatic rotation to keep it secure.



WHY THIS IS THE BEST WAY?

- No hardcoded credentials
- No secrets in Git or image
- Centralized, secure & auditable
- Automatic rotation support
- Highly available & scalable

COMMON MISTAKES TO AVOID

- Hardcoding secrets in code
- Storing secrets in Git
- Over-permissive IAM policies
- Not rotating secrets

SECURE YOUR SECRETS.
PROTECT YOUR APPLICATION.
BUILD WITH CONFIDENCE.

SAVE THIS SLIDE | SHARE IT | FOLLOW FOR MORE CLOUD KNOWLEDGE!